

ACTRA

Autonomous Work Operating System

An AI worker that takes a high-level goal, performs real work through tools, adapts to failures, asks for approval when required, and verifies the outcome before claiming completion.

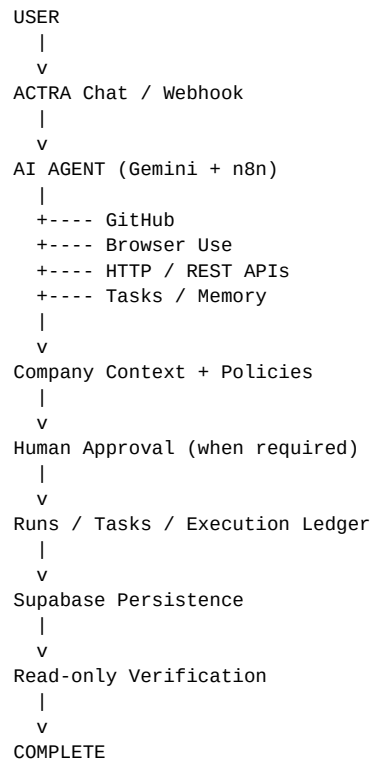
Project Overview

ACTRA is a prototype autonomous AI worker designed around outcome-driven execution rather than conversational assistance. A user provides a goal in natural language; ACTRA determines what needs to be done, selects tools, executes actions, observes results, recovers from failures, maintains execution state, and independently verifies the outcome.

Core loop: Understand → Plan → Execute → Observe → Adapt/Recover → Verify → Complete

The project focuses on autonomy, execution, reliability, recovery, verification, persistent state, human approval, and auditability.

Architecture



Core Capabilities

Capability	What it does
Goal understanding	Interprets high-level outcomes without requiring every step.
Planning & tool selection	Determines actions and selects appropriate tools.
Real execution	Uses GitHub, browser automation, APIs and internal tools.
Failure recovery	Detects failures and selects safe alternatives.
Human approval	Pauses before sensitive or irreversible actions.
Persistent state	Maintains one run_id and execution state.
Auditability	Records meaningful actions in an execution ledger.
Independent verification	Checks the resulting state before completion.
External persistence	Stores execution records in Supabase.
Resumability	Continues interrupted runs from recorded state.

Execution Model

Understand: interpret the intended outcome.

Plan: determine the minimum actions required.

Execute: select and call tools.

Observe: inspect actual results.

Adapt / Recover: handle failures safely.

Verify: independently check the resulting state.

Complete: report completion only with evidence.

Human-in-the-Loop

Low-risk work can be autonomous. External communications, financial transactions, deletion, permission changes, and other irreversible actions can require human approval.

Failure Recovery

```
Primary Tool → HTTP 500
↓
Observe Failure
↓
Safe Alternative
↓
Fallback Tool → HTTP 200
↓
Verify → RECOVERED
```

Verification

A successful tool call is not treated as proof of business success. A dedicated read-only verifier checks the exact run_id and returns NOT_VERIFIED if the requested record cannot be found.

Persistent State

ACTRA Runs: current execution state including run_id, goal, status, current_step, next_action, errors and result.

ACTRA Tasks: internal work generated during execution.

ACTRA Execution Ledger: meaningful execution events containing run_id, goal, action, tool, result and status.

Supabase: external persistence for execution records.

Resumability

If a run already exists, ACTRA can read its state, continue from the recorded step, avoid recreating completed work, and finish with independent verification.

Flagship Demonstration

“Prepare tomorrow's ACTRA product launch. Use the available company context and policies, inspect the GitHub repository, create the necessary internal launch tasks, persist the execution to Supabase, and independently verify the same record. Do not send external communications or make irreversible changes.”

Flow: Company Context → GitHub inspection → launch-readiness analysis → internal tasks → ledger → Supabase write → independent read-back → verified completion.

Technology Stack

Layer	Technology
LLM	Google Gemini
Agent / Runtime	n8n AI Agent + n8n workflow runtime
Browser	Browser Use
APIs	GitHub API, HTTP/REST, Supabase REST API
Persistence	n8n Data Tables + Supabase PostgreSQL
Memory / Approval	n8n Simple Memory + Human Review

Setup & Run

Requirements: Node.js 22+, n8n, Gemini API access, GitHub API access, Browser Use access, and a Supabase project.

```
nvm use 22
n8n
```

```
Open: http://localhost:5678
```

Import the sanitized workflow, configure your own credentials, publish/activate it, then submit a natural-language goal through the chat interface or webhook.

Security: Never commit API keys, OAuth tokens, Supabase secret/service-role keys, .env files, private credentials, or confidential information.

Known Limitations

This is a prototype. Limitations include a small integration set, dependence on LLM behavior, n8n-based orchestration, limited distributed execution and concurrency, browser sensitivity to website changes, additional work needed for binary document processing, and the need for stronger production authorization, security, observability and automated evaluation.

What I Would Build Next

With another two weeks, I would build a dedicated durable execution layer with queues, checkpoints, idempotency and distributed workers; add deterministic policy enforcement and stronger security; build an automated evaluation suite; improve observability; and make one end-to-end business workflow production-like reliable.

Design Philosophy

An AI worker should be responsible for the outcome, not merely the response.

ACTRA is therefore designed around execution, state, recovery, verification and controlled autonomy.